

Homework — 1.1.1 Structure and Function of the Processor — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Part A (14 marks)

1. [4] — 2 correct cells per mark. Completed table:

Register	Full name	Role
PC	Program Counter	Holds the address of the next instruction to be fetched (incremented each cycle)
ACC	Accumulator	Stores the immediate result of ALU calculations
MAR	Memory Address Register	Holds the address of the location in memory to be read from / written to
CIR	Current Instruction Register	Holds the instruction currently being decoded and executed

2. [2] — Correct order:

Event	Order
Instruction transferred from memory to the MDR along the data bus	2
CU decodes the instruction in the CIR into opcode and operand	5
Address held in the PC is copied into the MAR	1
Contents of the MDR are copied into the CIR	4
PC is incremented by 1	3

All five correctly placed = 2 marks; three or four correctly placed = 1 mark. (Accept the PC increment at position 2, 3 or 4 provided the other events keep their relative order.)

3. [3] — 1 mark per fully correct row:

PC	CIR	ACC	Contents of address 6
1	LOAD 4	7	0
2	ADD 5	12	0
3	STORE 6	12	12

(Working: LOAD 4 copies 7 into ACC; ADD 5 gives $7 + 5 = 12$; STORE 6 writes 12 to address 6. The PC is incremented during each fetch, so it reads 1, 2, 3.)

4. [3] — 2 correct blanks per mark: (a) **clock**; (b) **cache**; (c) **RAM**; (d) **pipelining**; (e) **executed**; (f) **core**. (Distractors not used: interrupt, register, ROM.)

5.

(a) [1] — $2^{16} = 65,536$ locations. (Accept "64K".)

(b) [1] — More bits are carried per transfer, so more data/instructions move per cycle → higher throughput / faster data movement.

Part B (6 marks)

6. [3] — 1 mark per row: Address bus = **B**; Data bus = **A**; Control bus = **C**. (*D and E are distractors.*)

7. [2] — 1 mark: cache is **small, very fast memory on/near the CPU** holding frequently/recently used data/instructions (not part of RAM). 1 mark: it reduces the need to fetch from **slower RAM**, so data is retrieved faster / fewer slow main-memory accesses.

8. [1] — Pipelining = while one instruction is being **executed**, the next is being **decoded** and a third **fetches** — the stages of the FDE cycle **overlap** for consecutive instructions, keeping all parts of the processor busy / increasing throughput.

Total: 14 + 6 = 20